Boosting

Gradient Boosting: Advanced CWB

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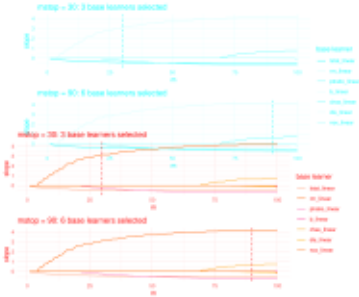

Learning goals

- Details of nonlinear BLs and splines
- Decomposition for splines
- Fair base learner selection
- Feature importance and PDPs

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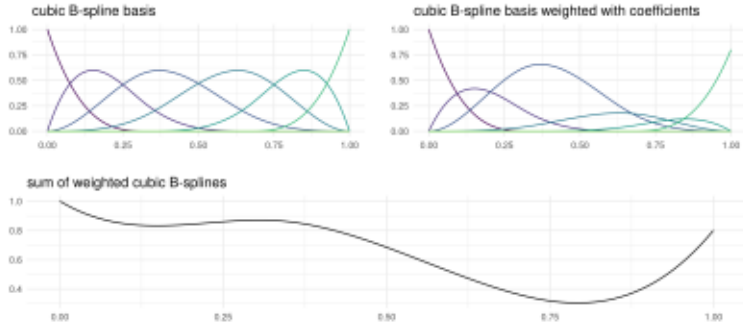
NONLINEAR BASE LEARNERS

As an alternative we can use nonlinear base learners, such as *P*- or *B*-splines, which make the model equivalent to a **generalized additive model (GAM)** (as long as the base learners keep their additive structure, which is the case for splines).



NONLINEAR EFFECT DECOMPOSITION

▸ Kneib, Hothorn and Tutz (2009) proposed a decomposition of each base learner into a constant, a linear and a nonlinear part. The boosting algorithm will automatically decide which feature to include linear, nonlinear, or none at all:

$$
\begin{aligned}
b_j(x_j, \boldsymbol{\theta}^{[m]}) &= b_{j,\text{const}}(x_j, \boldsymbol{\theta}^{[m]}) + b_{j,\text{lin}}(x_j, \boldsymbol{\theta}^{[m]}) + b_{j,\text{nonlin}}(x_j, \boldsymbol{\theta}^{[m]}) \\
&= \theta^{[m]}_{j,\text{const}} + x_j \cdot \theta^{[m]}_{j,\text{lin}} + s_j(x_j, \boldsymbol{\theta}^{[m]}_{j,\text{nonlin}}),
\end{aligned}
$$

where

- $\theta_{j,\text{const}}$ is the intercept of feature j,
- $x_j \cdot \theta^{[m]}_{j,\text{lin}}$ is a feature-specific linear base learner, and
- $s_j(x_j, \boldsymbol{\theta}^{[m]}_{j,\text{nonlin}})$ is a (centered) nonlinear base learner capturing deviation from the linear effect

Careful: We usually also apply an orthogonalization procedure on top of this but skip technical details here.

PARTIAL DEPENDENCE PLOTS

If we use single features in base learners, we consider each BL as a wrapper around a feature representing the feature's effect on the target. BLs can be selected more than once (with varying parameter estimates), signaling that this feature is more important.
E.g. let $j \in \{1, 2, 3\}$, the first three iterations might look as follows

$$
\begin{aligned}
m = 1: &\quad \hat{f}^{[1]}(\mathbf{x}) = \hat{f}^{[0]} + \alpha \hat{b}_2(x_2, \hat{\theta}^{[1]}) \\
m = 2: &\quad \hat{f}^{[2]}(\mathbf{x}) = \hat{f}^{[1]} + \alpha \hat{b}_3(x_3, \hat{\theta}^{[2]}) \\
m = 3: &\quad \hat{f}^{[3]}(\mathbf{x}) = \hat{f}^{[2]} + \alpha \hat{b}_2(x_2, \hat{\theta}^{[3]})
\end{aligned}
$$

Due to linearity, $\hat{b}_2$ base learners can be aggregated:

$$\hat{f}^{[3]}(\mathbf{x}) = \hat{f}^{[0]} + \alpha(\hat{b}_2(x_2, \hat{\theta}^{[1]} + \hat{\theta}^{[3]}) + \hat{b}_3(x_3, \hat{\theta}^{[2]}))$$

Which is equivalent to: $\hat{f}^{[3]}(\mathbf{x}) = \hat{f}_0 + \hat{f}_2(x_2) + \hat{f}_3(x_3)$.
Hence, $\hat{f}$ can be decomposed into the marginal feature effects (PDPs).

FEATURE IMPORTANCE

- We can further exploit the additive structure of the boosted ensemble to compute measures of **variable importance**.
- To this end, we simply sum for each feature x_j the improvements in empirical risk achieved over all iterations until $1 < m_{\text{stop}} \leq M$:

$$VI_j = \sum_{m=1}^{m_{\text{stop}}} \left(\mathcal{R}_{\text{emp}} \left(f^{[m-1]}(\mathbf{x}) \right) - \mathcal{R}_{\text{emp}} \left(f^{[m]}(\mathbf{x}) \right) \right) \cdot \mathbb{I}_{[j \in j^{[m]})]},$$

TAKE-HOME MESSAGE

- Componentwise gradient boosting is the statistical re-interpretation of gradient boosting
- We can fit a large number of statistical models, even in high dimensions ($p \gg n$)
- A drawback compared to statistical models is that we do not get valid inference for coefficients $\rightarrow$ post-selection inference
- In most cases, gradient boosting with trees will dominate componentwise boosting in terms of performance due to its inherent ability to include higher-order interaction terms